

Math 5023
Homework 2

Name: Gianluca Crescenzo

I skipped problem 8 because I had Covid this week and did not manage my time properly enough.

Exercise 1. For an R -module M , let $\text{End}_R(M) = \text{Hom}_R(M, M)$, the R -module of *endomorphisms*. We have seen that it is a ring. Compute $\text{End}_R(M)$ when $M = R$ and $M = R/I$ for an ideal I .

Proof. Claim: $\text{End}_R(R/I) \cong R/I$. Define $\varphi : R \rightarrow \text{End}_R(R/I)$ by $r \mapsto L_{r+I}$, where $L_{r+I} : R/I \rightarrow R/I$ is left multiplication by $r + I$. For $r_1, r_2 \in R$ we have:

$$\begin{aligned}\varphi(r_1 + r_2)(x + I) &= L_{r_1 + r_2 + I}(x + I) \\ &= (r_1 + r_2)x + I \\ &= (r_1x + I) + (r_2x + I) \\ &= \varphi(r_1)(x + I) + \varphi(r_2)(x + I) \\ &= (\varphi(r_1) + \varphi(r_2))(x + I).\end{aligned}$$

For $r \in R$ and $a \in R$ we have:

$$\begin{aligned}\varphi(ar)(x + I) &= L_{ar+I}(x + I) \\ &= arx + I \\ &= a(rx + I) \\ &= a\varphi(r)(x + I).\end{aligned}$$

Thus φ is an R -module homomorphism. By the First Isomorphism Theorem for Modules, $R/\ker \varphi \cong \varphi(R)$. For any $f \in \text{End}_R(R/I)$, note that

$$f(x + I) = f(1 + I) \cdot x = L_{f(1+I)}(x + I),$$

Hence $\varphi(f(1 + I)) = f$. Similarly:

$$\begin{aligned}\ker \varphi &= \{r \in R : \varphi(r)(x + I) = 0 + I \text{ for all } x \in R\} \\ &= \{r \in R : L_{r+I}(x + I) = 0 + I \text{ for all } x \in R\} \\ &= \{r \in R : rx \in I \text{ for all } x \in R\} \\ &= I.\end{aligned}$$

Thus $R/I \cong \text{End}_R(R/I)$. Taking $I = \{0\}$, we also have $R \cong \text{End}_R(R)$. ■

Exercise 2. We say an R -module has *rank one* if it has a linearly independent generating set consisting of a single element. Any ring R with 1 can be viewed as an R -module, namely the R -module $\langle 1_R \rangle$. We say it is a “module over itself”. Consider $R = \mathbb{R}[x, y]$, viewed as a module over itself. Prove that it is free of rank one, but that the submodule $I = (x, y)$ is not free, and needs at least two generators.

Proof. Since $\langle 1_{R[x,y]} \rangle = R[x, y]$, $R[x, y]$ is rank one with basis $\{1_{R[x,y]}\}$. This is indeed a basis, as it cannot be linearly dependent. Hence it is free of rank one. Moreover, the submodule $I = (x, y)$ is not principal, hence it cannot be generated by a single element. It remains to show that I is not free. Note that $\{x, y\}$ is not linearly independent as $(-y)x + (x)y = 0$. Thus I is not free. ■

Exercise 3. Let $I \subset R$ be an ideal. Prove that I is a free R -module if and only if I is principal that is generated by a non zero-divisor.

Proof. ($\Rightarrow$) Let I be a free module. Claim: I is rank 1. Indeed, if I had e_1 and e_2 as basis elements, then $e_2e_1 - e_1e_2 = 0$ by the commutativity of R . This means e_1 and e_2 are linearly dependent, a contradiction. Since I is rank 1, $I = (e)$ for some $e \in R$. Moreover, e is not a zero divisor, because if it were then there would exist an element $r \in R$, $r \neq 0$ such that $re = 0$. This means e is linearly dependent, again contradicting the fact that I is free. Thus I is a principal ideal generated by a non zero-divisor.

($\Leftarrow$) Suppose $I = (e)$ where e is not a zero divisor. Then e is linearly independent, hence I is free of rank 1. ■

Exercise 4. An idempotent e of a ring R (with 1) is any element satisfying $e^2 = e$. We usually assume an idempotent is nontrivial, meaning $e \neq 0, 1$. Suppose R has a nontrivial idempotent e . Prove that there is an R -module isomorphism $Re \oplus R(1 - e) \simeq R$.

Proof. Define $\varphi : R \rightarrow Re \oplus R(1 - e)$ by $r \mapsto re + r(1 - e)$. For $r_1, r_2 \in R$ we have:

$$\begin{aligned}\varphi(r_1 + r_2) &= (r_1 + r_2)e + (r_1 + r_2)(1 - e) \\ &= r_1e + r_1(1 - e) + r_2e + r_2(1 - e) \\ &= \varphi(r_1) + \varphi(r_2).\end{aligned}$$

Similarly, for $r, a \in R$:

$$\begin{aligned}\varphi(ar) &= are + ar(1 - e) \\ &= a(re + r(1 - e)) \\ &= a\varphi(r).\end{aligned}$$

Thus φ is an R -module homomorphism. For $r_1, r_2 \in R$, suppose $\varphi(r_1) = \varphi(r_2)$. Then $r_1e + r_1(1 - e) = r_2e + r_2(1 - e)$, which is equivalent to $r_1 = r_2$. Thus φ is injective. Suppose now that $x \in Re \oplus R(1 - e)$. Then $x = a + b$, where $a \in Re$ and $b \in R(1 - e)$. Then $\varphi(a + b) = \varphi(a) + \varphi(b) = ae + a(1 - e) + be + b(1 - e) = a + b = x$. Thus φ is surjective, which means $R \cong Re \oplus R(1 - e)$. ■

Exercise 5. Let A, B, M, N be R -modules. Use the universal properties of the coproduct and product to prove:

- (a) There is a natural isomorphism $\text{Hom}_R(A \oplus B, M) \longrightarrow \text{Hom}_R(A, M) \times \text{Hom}_R(B, M)$.
- (b) There is a natural isomorphism $\text{Hom}_R(A, M \times N) \longrightarrow \text{Hom}_R(A, M) \times \text{Hom}_R(A, N)$.

Proof. (a) Since coproducts exist in the category of R -modules, let $\iota_A : A \rightarrow A \oplus B$ and $\iota_B : B \rightarrow A \oplus B$ be the natural inclusions. Define $\varphi : \text{Hom}_R(A \oplus B, M) \rightarrow \text{Hom}_R(A, M) \times \text{Hom}_R(B, M)$ by $\varphi(\tau) := (\tau \circ \iota_A, \tau \circ \iota_B)$. For $\tau_1, \tau_2 \in \text{Hom}_R(A \oplus B, M)$, we have:

$$\begin{aligned}\varphi(\tau_1 + \tau_2) &= ((\tau_1 + \tau_2) \circ \iota_A, (\tau_1 + \tau_2) \circ \iota_B) \\ &= (\tau_1 \circ \iota_A, \tau_1 \circ \iota_B) + (\tau_2 \circ \iota_A, \tau_2 \circ \iota_B) \\ &= \varphi(\tau_1) + \varphi(\tau_2).\end{aligned}$$

For $\tau \in \text{Hom}_R(A \oplus B, M)$ and $r \in R$ we have:

$$\begin{aligned}\varphi(r\tau) &= ((r\tau) \circ \iota_A, (r\tau) \circ \iota_B) \\ &= r(\tau \circ \iota_A, \tau \circ \iota_B) \\ &= r\varphi(\tau).\end{aligned}$$

Thus φ is an R -module homomorphism. It remains to show that φ is injective and surjective. For $\tau_1, \tau_2 \in \text{Hom}_R(A \oplus B, M)$, suppose $\varphi(\tau_1) = \varphi(\tau_2)$. Then $(\tau_1 \circ \iota_A, \tau_1 \circ \iota_B) = (\tau_2 \circ \iota_A, \tau_2 \circ \iota_B)$. Since ι_A and ι_B are injective, we have that $\tau_1 = \tau_2$, implying that φ is injective. Now let $(f, g) \in \text{Hom}_R(A, M) \times \text{Hom}_R(B, M)$. By the Universal Property of Coproducts, there exists a map $\tau : A \oplus B \rightarrow M$ such that $f = \tau \circ \iota_A$ and $g = \tau \circ \iota_B$. Thus $\varphi(\tau) = (\tau \circ \iota_A, \tau \circ \iota_B) = (f, g)$.

(b) Since products exist in the category of R -modules, let $\pi_A : A \times B \rightarrow A$ and $\pi_B : A \times B \rightarrow B$ be the natural projections. Define $\psi : \text{Hom}_R(A, M \times N) \rightarrow \text{Hom}_R(A, M) \times \text{Hom}_R(A, N)$ by $\psi(f) = (\pi_M \circ f, \pi_N \circ f)$. For $f_1, f_2 \in \text{Hom}_R(A, M \times N)$, we have:

$$\begin{aligned}\psi(f_1 + f_2) &= (\pi_M \circ (f_1 + f_2), \pi_N \circ (f_1 + f_2)) \\ &= (\pi_M \circ f_1, \pi_N \circ f_1) + (\pi_M \circ f_2, \pi_N \circ f_2) \\ &= \psi(f_1) + \psi(f_2).\end{aligned}$$

For $f \in \text{Hom}_R(A, M \times N)$ and $r \in R$, we have:

$$\begin{aligned}\psi(rf) &= (\pi_M \circ (rf), \pi_N \circ (rf)) \\ &= r(\pi_M \circ f, \pi_N \circ f) \\ &= r\psi(f).\end{aligned}$$

Thus ψ is an R -module homomorphism. It remains to show that ψ is injective and surjective. For $f_1, f_2 \in \text{Hom}_R(A, M \times N)$, suppose $\psi(f_1) = \psi(f_2)$. Then $\pi_M \circ f_1 = \pi_M \circ f_2$ and $\pi_N \circ f_1 = \pi_N \circ f_2$. By the Universal Property of Products, there is a unique map from A to $M \times N$ with these two compositions. Hence $f_1 = f_2$, implying that ψ is injective. Now let $(g, h) \in \text{Hom}_R(A, M) \times \text{Hom}_R(A, N)$. By the Universal Property of Products, there exists a map $f : A \rightarrow M \times N$ such that $g = \pi_M \circ f$ and $h = \pi_N \circ f$. Thus $\psi(f) = (\pi_M \circ f, \pi_N \circ f) = (g, h)$. Therefore ψ is surjective, and hence an isomorphism. ■

Exercise 6. Use universal properties to show that there is a canonical morphism $\theta : \coprod_I M_i \rightarrow \prod_I M_i$, defined by $\theta(\sum m_i) = (m_i)$. You should use the fact that in the category of R -modules there is a zero object.

Proof. Since the category of R -modules admits a zero object, from the following diagram:

$$M_i \longrightarrow 0 \longrightarrow M_j$$

we see that there is a canonical zero morphism between any two modules. For each $j \in I$, define $\delta_j : M_j \rightarrow \prod_I M_i$ by:

$$\delta_j(x) = \begin{cases} x, & i = j \\ 0, & i \neq j \end{cases}.$$

By the Universal Property of Coproducts, there exists a unique map $\theta : \coprod_I M_i \rightarrow \prod_I M_i$ such that the following diagram commutes:

$$\begin{array}{ccc} & & \prod_I M_i \\ & \nearrow \theta & \uparrow \iota_j \\ \prod_I M_i & \xleftarrow{\delta_j} & M_j \end{array}$$

where ι_j is the natural inclusion. ■

Exercise 7. Let R be a commutative ring with 1, and suppose we have a short exact sequence of R -modules. A *short exact sequence* of R -modules is a diagram of the form

$$1 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 1$$

such that the kernel of each map equals the image of the preceding map, where applicable. Thus ι is injective, π is surjective, and $\iota(M') = \ker(\pi)$, so that $M/\iota(M') \simeq M''$. We say the sequence *splits* if there is a morphism $\sigma : M'' \longrightarrow M$ such that $\pi \circ \sigma = \text{id}_{M''}$. Use the universal property of free modules to prove that if M'' is free then the sequence splits, and then $M \simeq M' \oplus M''$.

Proof. Since M'' is free, let $\{x_i\}_I$ be its basis. Let $e : \{x_i\}_I \rightarrow M''$ be the natural inclusion. Since π is surjective, pick $\{m_i\}_I \subset M$ such that $\pi(m_i) = x_i$ for each $i \in I$. Define $f : \{x_i\}_I \rightarrow M$ by $f(x_i) := \pi^{-1}(e(x_i)) = m_i$. By the Universal Property of Free Products, there exists a unique R -linear map $\sigma : M'' \rightarrow M$ such that $f = \sigma \circ e$. Thus:

$$\begin{aligned} \pi(\sigma(x_i)) &= \pi(\sigma(e(x_i))) \\ &= \pi(f(x_i)) \\ &= \pi(m_i) \\ &= x_i. \end{aligned}$$

Since $\pi \circ \sigma$ is the identity on the basis elements of M'' , we can conclude that $\pi \circ \sigma = \text{id}_{M''}$.

Now define $\varphi : M' \oplus M'' \rightarrow M$ by $m' + m'' \mapsto \iota(m') + \sigma(m'')$. Clearly this map is R -linear. To show that the map is injective, suppose $\varphi(m' + m'') = 0$. Then $\iota(m') + \sigma(m'') = 0$. Apply π to both sides yields $\pi(\iota(m')) + \pi(\sigma(m'')) = 0$. By exactness, $\pi(\iota(m')) = 0$. Since our sequence splits, $\pi(\sigma(m'')) = m''$. Thus $m'' = 0$. Hence $\iota(m') = 0$, and because ι is injective it must be that $m' = 0$. Thus $m' + m'' = 0$, establishing that φ is injective. Now let $m \in M$. Note that $\pi(m - \sigma(\pi(m))) = 0$, hence $m - \sigma(\pi(m)) \in \ker \pi$. Exactness of our sequence gives $\ker \pi = \iota(M')$. So there exists some $m' \in M'$ such that $m - \sigma(\pi(m)) = \iota(m')$. Thus $m = \iota(m') + \sigma(\pi(m)) = \varphi(m', \pi(m))$. Thus $M \cong M' \oplus M''$. ■

Exercise 8. Let R be a commutative ring with 1. Suppose we have a commutative diagram of R -modules

$$\begin{array}{ccccccc} 0 & \longrightarrow & M' & \longrightarrow & M & \xrightarrow{f} & M'' \longrightarrow 0 \\ & & & & \downarrow \wr & & \downarrow \wr \\ 0 & \longrightarrow & N' & \longrightarrow & N & \xrightarrow{g} & N'' \longrightarrow 0 \end{array}$$

Prove that there is an isomorphism $M' \longrightarrow N'$ that makes the resulting left square commutative.